

Tier 1 Minimal Model — Basketball Application

Core Structural Insight

Basketball represents an especially clean application of the framework because:

selection pool is global while **opportunity pool is local**

The NBA draft selects globally across all eligible players, but opportunities to generate draft-relevant signals are allocated locally through: - team membership, - minutes, - role, - usage, - and coaching decisions.

Core Tier 1 Logic

The minimal proposed mechanism is:

heterogeneous local pools + **finite global selection capacity** + **local opportunity competition** → **nonlinear**

The goal of Tier 1 is to determine whether these minimal ingredients alone can generate the observed inverted-U relationship.

Basic Structure

Let player i belong to team j during season t .

Latent Ability

$$A_i$$

where: - A_i = latent player ability.

In empirical applications, the observable own-performance metric (OPM) may serve as a proxy for latent ability A_i .

Examples: - basketball: points per minute (PPM), - Army: TB ratio / evaluation quality, - academia: publications, citations, or productivity measures.

Throughout discussion, "OPM" may therefore refer to the domain-specific empirical proxy used to operationalize latent ability.

Local Pool Quality

Define leave-self-out teammate quality:

$$Q_{jt}^{(-i)} = \frac{1}{|P_{jt}| - 1} \sum_{\ell \in P_{jt}, \ell \neq i} A_{\ell}$$

where: - P_{jt} = team roster / local opportunity pool, - $Q_{jt}^{(-i)}$ = average teammate quality excluding ego.

Observed Signal

Observed signal is generated through both: - individual ability, - and local environmental effects.

$$S_{ijt} = A_i + B(Q_{jt}^{(-i)}) + \varepsilon_{ijt}$$

where: - S_{ijt} = observed player signal, - $B(Q)$ = developmental / signaling benefit from stronger teammates, - ε_{ijt} = stochastic noise term.

Local Opportunity Allocation

Opportunities are allocated locally.

$$O_{ijt} = O(A_i, Q_{jt}^{(-i)}, C_{jt})$$

where: - O_{ijt} = opportunity allocation, - examples include: - minutes, - usage, - touches, - offensive role, - visibility, - C_{jt} = local congestion / competition intensity.

Draft-Relevant Realized Signal

Realized draft signal depends on both: - observed ability signal, - and opportunity exposure.

$$R_{ijt} = S_{ijt} \cdot O_{ijt}$$

where: - R_{ijt} = realized draft-relevant signal.

Global Selection

Define:

$$\Lambda_t = \text{annual global selection capacity}$$

For the NBA draft:

$$\Lambda_t \approx 60$$

Draft outcome:

$$Y_i = 1$$

if player i is selected among the top Λ_t players globally.

Potential probabilistic formulation:

$$P(Y_i = 1) = \text{logit}^{-1} \left(\alpha + \beta_1 A_i + \beta_2 Q_{jt}^{(-i)} - \beta_3 C_{jt} \right)$$

Source of the Inverted-U

The central tradeoff:

$$\frac{\partial B(Q)}{\partial Q} > 0$$

but:

$$\frac{\partial O}{\partial Q} < 0$$

at sufficiently high local pool quality.

Interpretation: - stronger environments may improve development and signaling, - but concentrated talent may reduce opportunity share.

Therefore:

$$\frac{\partial P(Y_i = 1)}{\partial Q} = \text{development/signaling gains} - \text{opportunity congestion costs}$$

Potential turning point:

$$B'(Q^*) = G'(Q^*)$$

where: - Q^* = optimal local pool quality, - $G(Q)$ = congestion/opportunity-cost function.

Variable Glossary / Model Ontology

Symbol	Type	Meaning	Formulation / Construction
i	index	player	row-level player identifier
j	index	team	team identifier
t	index	season	season identifier
A_i or A_{ijt}	latent / proxied	player ability or own performance	empirical proxy varies by domain; basketball: PPM_{ijt}
P_{jt}	set	local opportunity pool	players on team j in season t
$Q_{jt}^{(-i)}$	constructed variable	leave-self-out local pool quality	$\frac{1}{ P_{jt} -1} \sum_{\ell \in P_{jt}, \ell \neq i} A_{\ell jt}$
S_{ijt}	latent signal	observed/evaluated player signal	$A_i + B(Q_{jt}^{(-i)}) + \varepsilon_{ijt}$
$B(Q)$	function	developmental / signaling benefit from local environment	increasing benefit term; candidate: $B'(Q) > 0$
ε_{ijt}	noise term	stochastic signal noise	distribution TBD
O_{ijt}	observed / constructed	locally allocated opportunity	minutes, usage, role, touches, visibility
C_{ijt}	constructed variable family	local congestion / roster pressure	candidate forms: C_{ijt}^τ , C_{ijt}^{sum}
C_{ijt}^τ	constructed variable	threshold pressure	$\sum_{\ell \in P_{jt}, \ell \neq i} \mathbf{1}(A_{\ell jt} > \tau)$
C_{ijt}^{sum}	constructed variable	total roster pressure / talent mass	$\sum_{\ell \in P_{jt}, \ell \neq i} A_{\ell jt}$
R_{ijt}	constructed variable	realized draft-relevant signal	$S_{ijt} \cdot O_{ijt}$
Λ_t	parameter	annual global selection capacity	NBA draft: $\Lambda_t \approx 60$
Y_i	outcome	draft outcome	1 if eventually drafted, else 0

Current Empirical Mapping

Current empirical operationalization:

Concept	Current Empirical Variable
individual ability proxy	points per minute (PPM)
local pool quality	leave-self-out teammate PPM
success event	eventual NBA draft selection

Important Conceptual Clarification

This framework distinguishes:

Structure	Basketball Example
global selection pool	NBA draft
local opportunity pool	team roster
local opportunity allocation	minutes, role, usage
global advancement outcome	draft selection

This distinction may become one of the central conceptual foundations of the broader theory.

Addendum — Candidate Definitions of Local Congestion / Roster Pressure

An important emerging distinction is:

$$Q_{jt}^{(-i)} \neq C_{ijt}$$

where: - $Q_{jt}^{(-i)}$ captures average local pool quality, - while C_{ijt} captures local crowding pressure / roster pressure.

This distinction may become theoretically important.

Examples: - a team may have moderate average talent but very high crowding pressure, - or a team may have high average talent concentrated in only one or two players.

Thus: - average environment quality, - and competition for scarce opportunity,
may represent distinct mechanisms.

Candidate Interpretation of C_{ijt}

Conceptually:

$$C_{ijt} = \text{competition for scarce local opportunity}$$

Examples: - minutes, - usage, - touches, - command opportunities, - authorship prominence, - evaluator attention.

Candidate Formulation 1 — Threshold Pressure

$$C_{ijt}^{\tau} = \sum_{\ell \in P_{jt}, \ell \neq i} \mathbf{1}(A_{\ell jt} > \tau)$$

Interpretation: - number of teammates above some meaningful performance threshold.

Potential strengths: - interpretable, - threshold can be tuned empirically, - captures “how many legitimate competitors surround ego?”

Potential limitation: - does not capture intensity above threshold.

Candidate Formulation 2 — Total Roster Pressure

$$C_{ijt}^{sum} = \sum_{\ell \in P_{jt}, \ell \neq i} A_{\ell jt}$$

Interpretation: - total local performance pressure generated by surrounding teammates.

Potential strengths: - captures total competitive talent mass, - reflects cumulative crowding pressure, - avoids reducing congestion to simple averages.

Potential limitation: - may require normalization for roster size or opportunity supply.

Important Conceptual Distinction

$$Q_{jt}^{(-i)}$$

asks:

“How strong is the average local environment?”

while:

$$C_{ijt}$$

asks:

"How much local competition exists for scarce opportunity?"

These may produce substantially different dynamics.

Cross-Domain Interpretation

Domain	OPM	Local Pool Quality	Crowding / Pressure
Basketball	points per minute (empirical proxy for A)	mean teammate quality	count/sum of strong teammates
Army	TB ratio / evaluation quality	mean officer quality	concentration of strong officers in SR pool
Academia	publications / citations	mean department productivity	concentration of highly productive peers

Importantly, different congestion formulations may ultimately prove more appropriate in different domains.

The framework should therefore initially treat:

$$C_{ijt}$$

as a family of candidate congestion measures rather than a single finalized variable.

Current Direction

The immediate next goal is not full realism, but:

identifying the minimal mechanism set capable of reproducing the observed empirical shape.

The Tier 1 framework should therefore remain: - modular, - interpretable, - empirically testable, - and computationally simple enough for preliminary simulation and fitting.